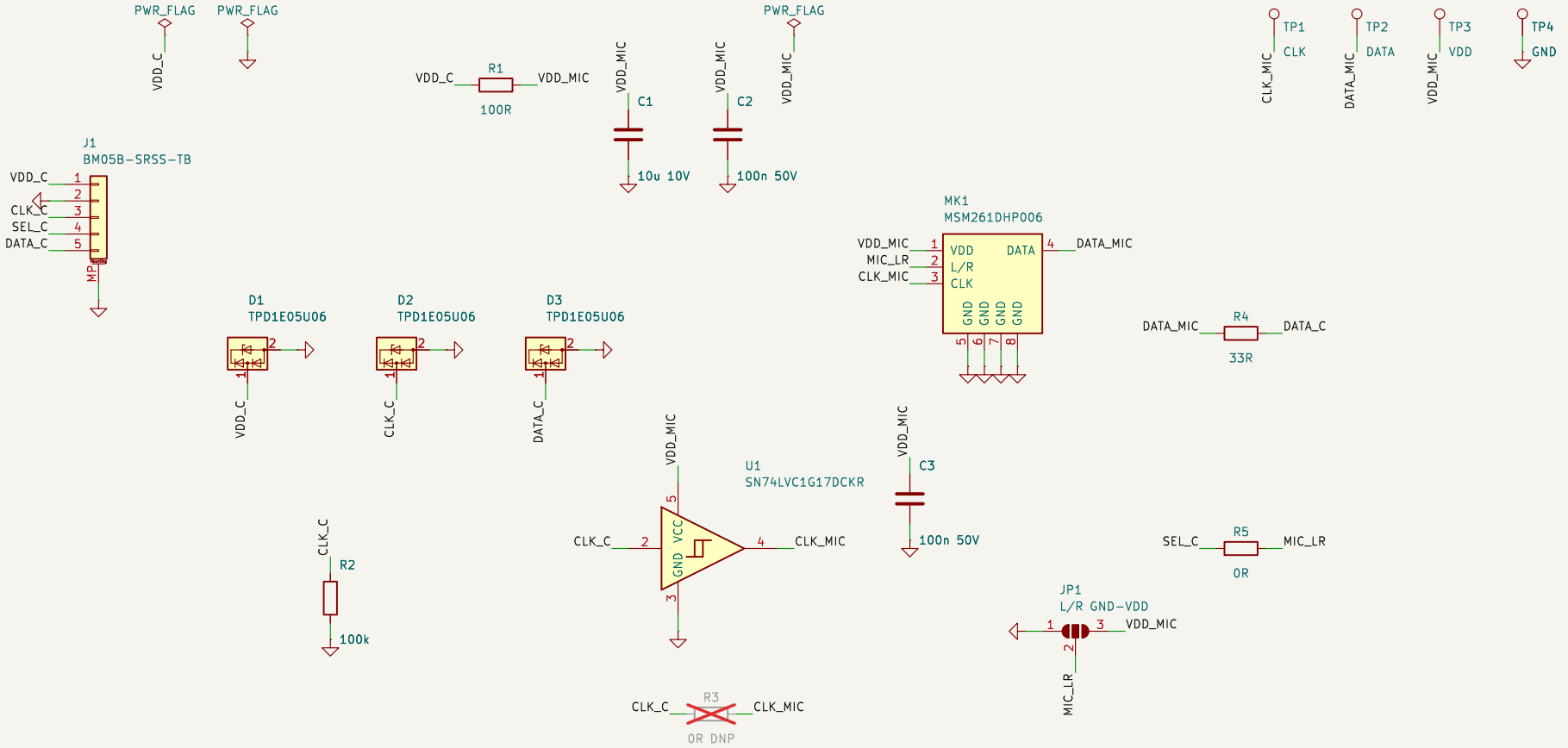
MIC STRIP (MSM261DHP006 PDM mic, one design for both strips; cable to the ReSpeaker Lite vacated U4/U5 pads)



MSM261DHP006 (MEMSensing DS-042 Ver.1.1): VDD 1.6-3.6 V, IDD 670 uA at 2.4 MHz, VIH 0.7 VDD, VIL 0.3 VDD, clock 1.1-4.8 MHz, edge <= 20 ns; L/R = GND -> DATA valid on CLK falling edge.
Cable (JST SR 1.0, A05SR05SR30K305B, cut >= 175 mm): 1 VDD, 2 GND, 3 CLK, 4 SEL, 5 DATA - SEL is static, so it separates CLK from DATA like a ground. Default: U1 fitted, R3 open, R4 33 Ohm, R5 0 Ohm, JP1 open.
Levers (bench 2.5): R3 bypass with U1 removed; R4 100 Ohm; R5 out + JP1 if the cable SEL is not wanted.
Passives 0603 (D42); ESD in SOD-523 (D42: leaded alternative to the X1SON); MK1 LGA accepted as is.

Two boards of this design: L and R set by the cable SEL wire
Brief claude/brief-electronics-design-2026-11-07.md, K21, E0-D6
KiCad 9.0.9 - generated by hardware/main-board/gen/mic_strip.py (checkpoint 1)
HA panel (neocleous/ha-panel)

Sheet: /
File: mic-strip.kicad_sch

Title: HA panel mic strip

Size: A4	Date: 2026-11-07	**Rev: 1**
KiCad E.D.A. 9.0.9		Id: 1/1